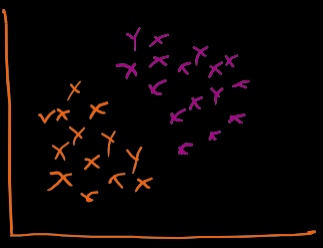
Thursday, October 15, 2026 11:37 AM

Naive Bayes

- It is a supervised ML algorithm
- used for only classification problems
- It is a probabistic algorithms.
- Base algo is Bayes Theorem.



purchased
not purchased

Based Age, Salary, ?.

Probability:

↳ chance.

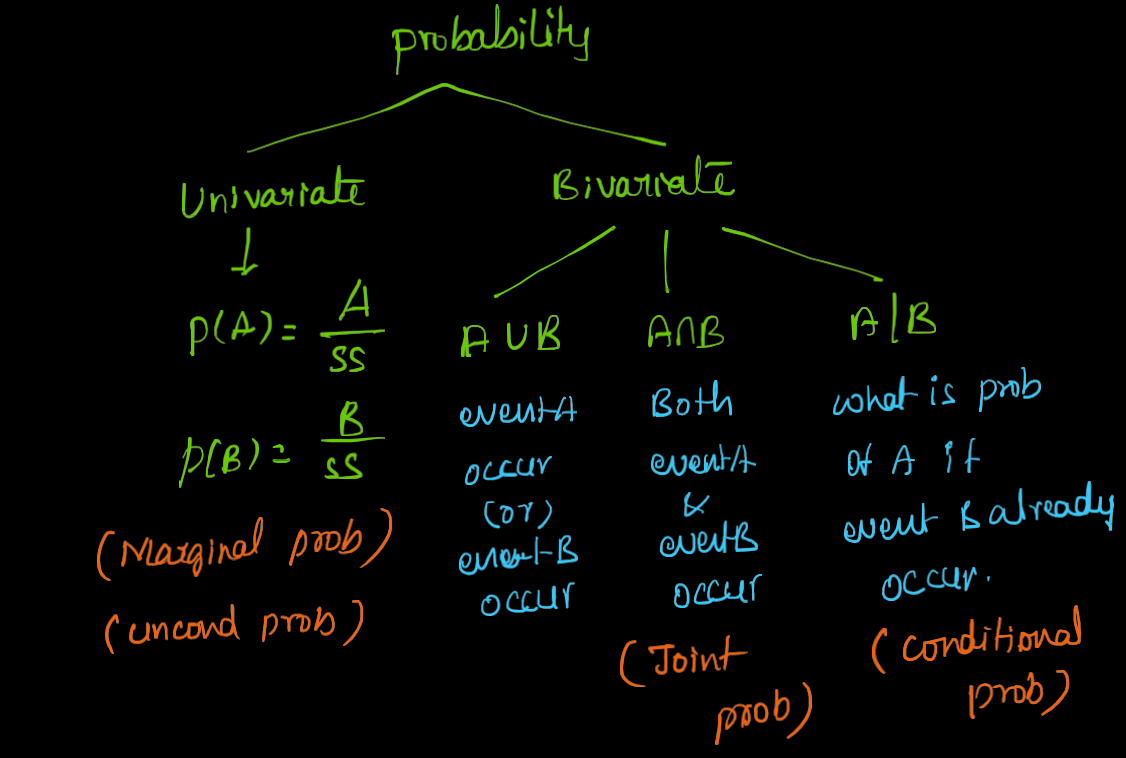
↓ → Random Experiment
↳ we can able to predict inprior

trail
↓
outcome
↓
sample space.
↳ event.

$$\text{probability} = \frac{\text{favourable cases}}{\text{Total no. of cases.}} \longrightarrow [0, 1]$$

Types of probabities:

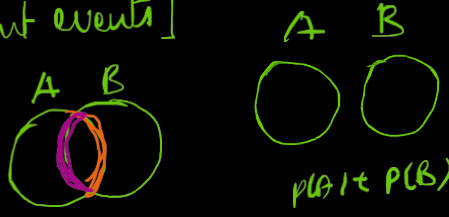
1. Conditional prob
2. Joint prob
3. Marginal prob.



$P(A \cup B) = P(A) + P(B)$ [Independant events]

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

[Dependant events]



$\cap \rightarrow$ AND $\rightarrow \times$

$\cup \rightarrow$ OR $\rightarrow +$

$P(A \cap B) = P(A) \cdot P(B)$ [Independant events]

$P(A \cap B) = P(A) \cdot P(B/A)$ [Dependant events]

$$P(B/A) = \frac{P(A \cap B)}{P(A)}$$

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

} → Dependant events

Independant event $P(A/B) = P(A)$

$P(B/A) = P(B)$

Bayes Theorem :

$$P(A/B) = \frac{P(B/A) \cdot P(A)}{P(B)}$$

As per cond prob $P(A/B) = \frac{P(A \cap B)}{P(B)} \Rightarrow P(A \cap B) = P(A/B) \cdot P(B)$

$P(B/A) = \frac{P(A \cap B)}{P(A)} \Rightarrow P(A \cap B) = P(B/A) \cdot P(A)$

$$P(A/B) \cdot P(B) = P(B/A) \cdot P(A)$$

$$\boxed{P(A/B) = \frac{P(B/A) \cdot P(A)}{P(B)}}$$

Age	Salary	purchased
25	50k	yes
35	40k	NO
:	:	:
40	60k	?

$P(Yes/age,sal)$

$P(NO/age,sal)$

$$P(A/B) = \frac{P(B/A) \cdot P(A)}{P(B)}$$

$$\Rightarrow P(Yes/age,sal) = \frac{P(age,sal/yes) \cdot P(Yes)}{P(age,sal)}$$

$$P(No/age, sal) = \frac{P(age, sal/No) \cdot P(No)}{P(age, sal)}$$

$P(yes/age, sal) = P(age, sal/yes) \cdot P(yes) = 0.7$
$P(No/age, sal) = P(age, sal/No) \cdot P(No) = 0.3$
} yes.

$P(C_1/x_1, x_2) = P(x_1, x_2/C_1) \cdot P(C_1)$

$P(C_2/x_1, x_2) = P(x_1, x_2/C_2) \cdot P(C_2)$

For multiclass & multi dimensional

C_k
x_n

$$P(C_k/x_1, x_2, x_3 \ldots x_n) = \frac{P(x_1, x_2, x_3 \ldots x_n/C_k) \cdot P(C_k)}{P(x_1, x_2, x_3 \ldots x_n)}$$

LHS RHS.

$\Rightarrow P(x_1, x_2, x_3, \ldots x_n/C_k) \cdot P(C_k)$

$\Rightarrow P(x_1 \cap x_2 \cap x_3 \cap \ldots x_n, C_k) \cdot P(C_k)$

$A \cap B$ $A \cup B$

AND OR

$x_1, x_2, x_3 \ldots x_n \rightarrow$ input variables.

$\Rightarrow P(\underbrace{x_1 \cap x_2 \cap x_3 \cap \ldots x_n}_{B}, C_k) \cdot P(C_k)$

A

$P(A \cap B) = P(A/B) \cdot P(B)$.

$$\Rightarrow p(x_1/C_k) \cdot p(\underbrace{x_2}_{A} \cap \underbrace{x_3 \cap x_4 \cap \dots x_n}_{B} / C_k) \cdot p(C_k)$$

$$\Rightarrow p(x_1/C_k) \cdot p(x_2/C_k) \cdot p(\underbrace{x_3}_{A} \cap \underbrace{x_4 \cap \dots x_n}_{B} / C_k) \cdot p(C_k)$$

$$\Rightarrow p(x_1/C_k) \cdot p(x_2/C_k) \cdot p(x_3/C_k) \cdot p(x_4 \cap x_5 \cap x_6 \dots x_n / C_k) \cdot p(C_k)$$

↓

Chain Rule in probability.

$$\Rightarrow p(x_1/C_k) \cdot p(x_2/C_k) \cdot p(x_3/C_k) \cdots p(x_n/C_k) \cdot p(C_k).$$

$$a_1 + a_2 + a_3 + \dots a_n \Rightarrow \sum_{i=1}^{n} a_i$$

$$a_1 \cdot a_2 \cdot a_3 \cdots a_n \Rightarrow \prod_{i=1}^{n} a_i$$

$$\Rightarrow \prod_{i=1}^{n} (x_i/C_k) \cdot p(C_k)$$

$$\therefore p(C_k/x_1, x_2, x_3 \dots x_n) = \frac{p(C_k) \prod_{i=1}^{n} (x_i/C_k)}{p(x_1, x_2, x_3 \dots x_n)}$$

0.0013

0.0259 + 1.0272.

0.0272.

$\frac{0.0013}{0.0272} = 0.0477$ (%).

$= 0.9497 \approx 1$.

95%.

Outlook	Temperature	Humidity	Windy	PlayTennis
Sunny 1	Hot	High	False 1	No
Sunny 2	Hot	High	True	No
Overcast	Hot	High	False 1	Yes
Rainy	Mild	High	False 2	Yes
Rainy	Cool 1	Normal 1	False 3	Yes
Rainy	Cool 1	Normal 1	True	No
Overcast	Cool 2	Normal 2	True	Yes
Sunny 3	Mild	High	False 2	No
Sunny 1	Cool 1	Normal 3	False 4	Yes
Rainy	Mild	Normal 4	False 5	Yes
Sunny 2	Mild	Normal 5	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal 6	False 6	Yes
Rainy	Mild	High	True	No

outlook Temp Hum Wind Play

sunny cool normal F ?

$p(yes) = \frac{9}{14}$, $p(no) = \frac{5}{14}$.

$p(yes | out=sunny, temp=cool, Hum=normal, win=F)$

$= p(sunny/yes) \cdot p(cool/yes) \cdot p(normal/yes) \cdot p(F/yes) \cdot p(yes)$

$p(no | out=sunny, temp=cool, Hum=normal, wid=F)$

$= p(sunny/no) \cdot p(cool/no) \cdot p(nor/no) \cdot p(F/no) \cdot p(no)$

$\Rightarrow \frac{2}{9} \times \frac{3}{9} \times \frac{6}{9} \times \frac{6}{9} \times \frac{9}{14}$

$= 0.3 \times 0.2 \times 0.6 \times 0.6 \times 0.64 = 0.013824$ ✓

$\Rightarrow \frac{3}{5} \times \frac{1}{5} \times \frac{1}{5} \times \frac{2}{5} \times \frac{5}{14}$

$= 0.6 \times 0.2 \times 0.2 \times 0.4 \times 0.35 = 0.0034$ ✓

$\Rightarrow \frac{0.0138}{0.0138+0.0034} + \frac{0.0034}{0.0138+0.0034}$

$\Rightarrow$ -

Hum Wind Play

outlook	Temp	Hum	Wind	Play
sunny	cool	normal	F	?
	moderate			

$$\Rightarrow \frac{3}{9} \times \frac{0}{9} \times \frac{2}{9} \cdots \Rightarrow 0.$$

To overcome the problem of zero / unseen data

Laplas Smoothening :

α = smoothen value.

k = no. of class.

N = no. of observation with resp to class

$$\alpha = 10^{-9} = 0.000000001.$$

$$\frac{p(x_i|c_k) + \alpha}{N + K\cdot\alpha}.$$

$$K = 2, N = 3, \alpha = 1$$

$$P(\text{sunny/yes}) = \frac{\frac{3}{9} + 1}{3+2\times1} = \frac{\frac{12+4}{9\times3}}{5} = \frac{4}{15} = 0.26$$

$$\frac{3}{9} = 0.33$$

$$P(\text{moderate/yes}) = \frac{\frac{0}{9} * 1}{3+2} = \frac{0+1}{\sim} \neq 0.\frac{1}{5} = 0.0005.$$

How to Handle numerical data ?

Age	purchase
35	no
45	yes
22	no
20	no
21	yes
15	yes
26	yes

- N.B will assume data will follows Gaussian distribution.

$$PDF = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot \exp\left(\frac{-(x-\mu)^2}{2\sigma^2}\right)$$

variance, mean

Gaussian NB ✓

Bernoulli NB ✓

Multinomial NB.

